# *Étale Coverings and the Fundamental Group*

(SGA 1)

English source-synchronization workpass

Exposé I opening through § I.3

Task-owned working translation

18 July 2026

## Source and status note

This bounded workpass translates the opening of Exposé I through § I.3 directly from the French TEX source in arXiv `math/0206203v2`, with archive SHA-256:

`0A33C9C06908705A2525690FAAE02F6F`
`07980A4AA069A8B6CFF9B1D9BC39ACD3`

The current translated source boundary is `smf_doc-math_3_01.tex`, lines 556–799; line 800 begins § I.4 and is outside the current workpass. The opening through § I.2 has passed a frozen checkpoint review; the bounded § I.3 addition has passed source, formula, structure, and independent text review.

Recovered English materials associated with jcreinhold, jmoellermath, and Tim Hosgood were consulted only as target-language comparison controls. They are not source authority, and no agreement among them was treated as proof of source alignment. The two Hosgood-related repositories belong to one derivative lineage, not two independent witnesses.

This is a source-critical working translation in bounded units. It is not a complete SGA 1 translation, a critical edition, or a certification of the mathematics. Rights and attribution for any public redistribution remain subject to archive-maintainer review; the French source header asks modified versions to identify themselves, which this workpass does explicitly.

# Contents

## CHAPTER I

# Étale morphisms

To simplify the exposition, we assume that all preschemes under consideration are locally noetherian, at least after no. I.2. French source p. 1

## I.1. Notions of differential calculus

Let $X$ be a prescheme over $Y$, and let $\Delta_{X/Y}$, or simply $\Delta$, be the diagonal morphism $X \to X \times_Y X$. It is an immersion, hence a closed immersion of $X$ into an open subset $V$ of $X \times_Y X$. Let $\mathcal{I}_X$ be the ideal of the closed subprescheme corresponding to the diagonal in $V$ (N.B. if one wishes to proceed intrinsically, without assuming that $X$ is separated over $Y$—a hypothesis that would be farcical—one should consider the set-theoretic inverse image of $\mathcal{O}_{X\times X}$ in $X$, and denote by $\mathcal{I}_X$ the augmentation ideal in the latter…). The sheaf $\mathcal{I}_X/\mathcal{I}_X^2$ may be regarded as a quasi-coherent sheaf on $X$; we denote it by $\Omega^1_{X/Y}$. It is of finite type if $X \to Y$ is of finite type. It behaves well under base change $Y' \to Y$. We also introduce the sheaves $\mathcal{O}_{X\times_Y X}/\mathcal{I}_X^{n+1} = \mathcal{P}^n_{X/Y}$; these are sheaves of *rings* on $X$, making $X$ into a prescheme that may be denoted by $\Delta^n_{X/Y}$ and called the *n-th infinitesimal neighborhood of* $X/Y$. The resulting sorites is entirely trivial, although rather long[1]; it would be prudent to speak of it only when there is something useful to say about it, in connection with smooth morphisms.

## I.2. Quasi-finite morphisms

PROPOSITION I.2.1. *Let $A \to B$ be a local homomorphism (N.B. the rings are now noetherian), and let $\mathfrak{m}$ be the maximal ideal of $A$. The following conditions are equivalent:*

(i) *$B/\mathfrak{m}B$ is finite-dimensional over $k = A/\mathfrak{m}$.*
(ii) *$\mathfrak{m}B$ is an ideal of definition, and $B/\mathfrak{r}(B) = \kappa(B)$ is an extension of $k = \kappa(A)$.*
(iii) *The completion $\widehat{B}$ is finite over the completion $\widehat{A}$ of $A$.*

One then says that $B$ *is quasi-finite* over $A$. A morphism $f\colon X \to Y$ is said to be quasi-finite at $x$ (or the $Y$-prescheme $f$ is said to be quasi-finite at $x$) if $\mathcal{O}_x$ is quasi-finite over $\mathcal{O}_{f(x)}$. This is also equivalent to saying that $x$ is *isolated in its fiber* $f^{-1}(f(x))$.[2] A morphism is said to be quasi-finite if it is so at every point.[3] French source p. 2

COROLLARY I.2.2. *If $A$ is complete, quasi-finite is equivalent to finite.*

One could give the usual chain of implications (i), (ii), (iii), (iv), (v) for quasi-finite morphisms, but this does not seem indispensable here.

## I.3. Unramified or net morphisms

PROPOSITION I.3.1. *Let $f\colon X \to Y$ be a morphism of finite type, let $x \in X$, and put $y = f(x)$. The following conditions are equivalent:*

[1] Cf. EGA IV 16.3.

[2] The original printing (Exposé I, p. 2) and both French arXiv renderings print $f^{-1}(x)$. Since $x \in X$ for $f\colon X \to Y$, the type-correct fiber through $x$ is $f^{-1}(f(x))$, which is used here with the source defect explicitly disclosed.

[3] In EGA II 6.2.3 one assumes in addition that $f$ is of finite type.

(i) $\mathcal{O}_x/\mathfrak{m}_y\mathcal{O}_x$ *is a finite separable extension of* $\kappa(y)$.
(ii) $\Omega^1_{X/Y}$ *vanishes at* $x$.
(iii) *The diagonal morphism* $\Delta_{X/Y}$ *is an open immersion in a neighborhood of* $x$.

For the implication (i)⇒(ii), Nakayama's lemma immediately reduces us to the case $Y = \operatorname{Spec}(k)$, $X = \operatorname{Spec}(k')$, where the assertion is well known and, moreover, immediate from the definition of separability. The implication (ii)⇒(iii) follows from a pleasant and easy characterization of open immersions using Krull. For (iii)⇒(i), one is again reduced to the case $Y = \operatorname{Spec}(k)$, with the diagonal morphism an open immersion everywhere. One must then prove that $X$ is finite, with a separable coordinate ring over $k$; for this one reduces to the case where $k$ is algebraically closed. But then every closed point of $X$ is isolated (it is precisely the inverse image of the diagonal under the morphism $X \to X \times_k X$ defined by $x$), and hence $X$ is finite. One may therefore suppose that $X$ consists of a single point, with ring $A$. Thus $A \otimes_k A \to A$ is an isomorphism, whence $A = k$, as required.

Definition I.3.2. (a) *One then says that* $f$ *is* net, *or equivalently* unramified, *at* $x$, *or that* $X$ *is net, or unramified, at* $x$ *over* $Y$.
(b) *Let* $A \to B$ *be a local homomorphism. It is said to be* net, *or* unramified, *and* $B$ *is said to be a net, or unramified, local algebra over* $A$, *if* $B/\mathfrak{r}(A)B$ *is a finite separable extension of* $A/\mathfrak{r}(A)$; *that is, if* $\mathfrak{r}(A)B = \mathfrak{r}(B)$ *and* $\kappa(B)$ *is a separable extension of* $\kappa(A)$.[4]

Remarks. The fact that $B$ is unramified over $A$ can already be checked on the completions of $A$ and $B$. Unramified implies quasi-finite. French source p. 3

Corollary I.3.3. *The set of points at which* $f$ *is unramified is open.*

Corollary I.3.4. *Let* $X'$ *and* $X$ *be two preschemes of finite type over* $Y$, *and let* $g\colon X' \to X$ *be a* $Y$*-morphism. If* $X$ *is unramified over* $Y$, *the graph morphism*

$$\Gamma_g\colon X' \longrightarrow X' \times_Y X$$

*is an open immersion.*[5]

Indeed, it is the inverse image of the diagonal morphism $X \to X \times_Y X$ under

$$g \times_Y \mathrm{id}_X\colon X' \times_Y X \longrightarrow X \times_Y X.$$

One may also introduce the annihilator ideal $\mathfrak{d}_{X/Y}$ of $\Omega^1_{X/Y}$, called the *different ideal* of $X/Y$. It defines a closed subprescheme of $X$ whose underlying set is the set of points at which $X/Y$ is ramified, that is, not unramified.

Proposition I.3.5. (i) *An immersion is unramified.*
(ii) *The composite of two unramified morphisms is unramified.*
(iii) *A base change of an unramified morphism is unramified.*

This may be seen equally well from criterion (ii) or criterion (iii) (the latter seems more amusing to me). One can of course also be more precise by giving pointwise statements; this is more general only in appearance (except in the setting of Definition I.3.2(b)), and tedious. As usual, one obtains the following corollaries.

Corollaries I.3.6. (iv) *The Cartesian product of two unramified morphisms is unramified.*
(v) *If* $gf$ *is unramified, then* $f$ *is unramified.*

[4] Cf. the regrets in Exposé III, no. 1.2.

[5] The original printing (Exposé I, p. 3) and the French TeX give $X \times_Y X$ as the graph target and later write $\mathrm{id}_{X'}$ in the pullback map. The typed fiber products force $X' \times_Y X$ and $\mathrm{id}_X$, respectively; those type-correct readings are used here with both source defects disclosed.

(vi) *If $f$ is unramified, then $f_{\mathrm{red}}$ is unramified.*

PROPOSITION I.3.7. *Let $A \to B$ be a local homomorphism. Suppose that the residue-field extension $\kappa(B)/\kappa(A)$ is trivial, or that $\kappa(A)$ is algebraically closed. Then $B/A$ is unramified if and only if $\widehat{B}$, as an $\widehat{A}$-algebra, is a quotient of $\widehat{A}$.*

Remarks.

- When the residue-field extension is not assumed to be trivial, one may reduce to that case by making a suitable finite flat extension over $A$ that kills the extension in question.
- Take the example in which $A$ is the local ring of an ordinary double point of a curve and $B$ is the local ring at a point of the normalization. Then $A \subset B$, and $B$ is unramified over $A$ with trivial residue-field extension; the map $\widehat{A} \to \widehat{B}$ is surjective but *not injective*. We shall therefore strengthen the notion of unramifiedness. French source p. 4